

# CS651 Project 1 Part 2

## Special Issues

### Special Issue 1: What happens when an EC2 instance stops running?

When an EC2 instance is stopped, anything stored on its ephemeral (instance-store) disks is lost. Our t3.micro uses EBS, so the 8 GB root volume actually persists across a stop/start -- but any data written outside EBS, and the running state (our Docker container!), does not survive: after a stop/start the container must be re-run, and the public IP changes (see Issue 2).

Solution -- "freeze" the instance as an AMI: In the EC2 console, select the instance > Actions > Image and templates > Create image. This snapshots the EBS volume into an AMI stored in S3. From that AMI you can launch an identical replacement instance at any time (same OS, Docker, and site files), or duplicate it for scaling. For data that changes often, snapshot the EBS volume on a schedule instead of relying on a one-time AMI.

### Special Issue 2: What happens on reboot -- the IP address changes

Every stop/start (and every reboot from a stopped state) assigns a new public IPv4 address, so a published URL like `http://100.25.220.164:3001` breaks. During our deployment the address changed three times across takes.

Options for a stable public URL:

- Elastic IP (what we used). Allocate an Elastic IP in the EC2 console and associate it with the instance. The address stays fixed across stop/start cycles, so the URL never breaks.
- DNS name. Point a domain (e.g. via Route 53) at the Elastic IP for a memorable URL instead of a raw IP.
- Load balancer. For production, an Application Load Balancer gives a fixed DNS name and health checks, at extra cost.

Fees: An Elastic IP attached to a running instance is free. It costs ~\$0.005/hour when attached to a stopped instance or left unattached, so release it when the project is done.

Learner Lab: Yes -- Elastic IPs can be allocated and associated inside the course Learner Lab (we did it in ours); the usual AWS limits apply.